

3.4 Summary guide to chart selection ↕

□ Watch: Selecting a chart type – part 3 [9.50 mins]

Watch the following video, which provides guidelines for selecting a chart type and explains why some chart types should be avoided.

This is useful as it enables a balance of understanding about the usefulness of different chart types.

[Transcript](#)

<https://learn.usc.edu.au/courses/35242/files/3138639?wrap=1>

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [Selecting a chart type – part 3](#)
<https://learn.usc.edu.au/courses/35242/files/3172991?wrap=1>

It is important to recognise that there are always exceptions to every rule. In general, the following applies to the chart or map described:

- Bar charts illustrate relationships between categories and the distribution of a variable over discrete categories or time periods. They are also useful for showing the composition of an entity, as they can indicate the variable's contribution to the whole.
- A bubble chart is useful when three quantitative variables are being compared, as it illustrates how the quantitative variable values are distributed over the range for each variable.

- Choropleth maps illustrate the distribution of quantitative or categorical data over a geographical area.
- Column charts can be used to illustrate relationships between categories and the distribution of a variable over discrete categories or time periods. Column charts are better for distribution over time.
- The rank order of categories on a quantitative variable can be visualised through sorted bar charts or column charts.
- A funnel chart can also demonstrate composition, as it can be used to drill down from the whole to individual variable (or the reverse).
- Heat maps illustrate relationships between categories.
- Line charts emphasise patterns across consecutive data points and trends over time.
- A scatter chart is useful for showing a relationship between two variables and for illustrating how the quantitative variable values are distributed over the range for each variable.
- Stock charts show relationships between time and stock price.
- A treemap is useful when illustrating the composition of the hierarchical structure between categorial variables.
- A waterfall chart can be used to visualise the composition of a quantitative variable over time or category.

□ Activity: Top 10 most valuable airline brands since 2013

Read and reflect on the following webpage about top-ranked airline brands.

As you read take time to consider:

- How easy is it for you to comprehend the first visualisation?
- Which of the data visualisations was easier for you to understand?

Explain your reasoning and bring your notes to the workshop.

Peasley, J. R. (2023, April 18). [Ranked: Top 10 most valuable airline brands since 2013](https://www.visualcapitalist.com/cp/ranked-top-10-most-valuable-airline-brands/)  [\(https://www.visualcapitalist.com/cp/ranked-top-10-most-valuable-airline-brands/\)](https://www.visualcapitalist.com/cp/ranked-top-10-most-valuable-airline-brands/). *Visual Capitalist*.

Chart types to avoid

Here are some chart types to avoid and alternative options to consider:

- *Pie charts*. People are better at assessing differences in length rather than angle and area, so use a bar chart instead.
- *Radar charts*. These can be difficult to interpret, so use a clustered column chart instead.

- *Area charts*. These can be cluttered and difficult to read, so use a line chart instead.
- *Combo charts*. These can be cluttered and difficult to interpret, so it's better to use separate charts.
- *3D charts*. Usually the third dimension only adds clutter to the chart without increasing understanding.